

Prevent Command History Logging

Adversaries may impair command history logging to hide commands they run on a compromised system. Various command interpreters keep track of the commands users type in their terminal so that users can retrace what they have done.

On Linux and macOS, command history is tracked in a file pointed to by the environment variable `HISTFILE`. When a user logs off a system, this information is flushed to a file in the user's home directory called `~/.bash_history`. The `HISTCONTROL` environment variable keeps track of what should be saved by the history command and eventually into the `~/.bash_history` file when a user logs out. `HISTCONTROL` does not exist by default on macOS, but can be set by the user and will be respected. The `HISTFILE` environment variable is also used in some ESXi systems.^[1]

Adversaries may clear the history environment variable (`unset HISTFILE`) or set the command history size to zero (`export HISTFILESIZE=0`) to prevent logging of commands. Additionally, `HISTCONTROL` can be configured to ignore commands that start with a space by simply setting it to "ignorespace". `HISTCONTROL` can also be set to ignore duplicate commands by setting it to "ignoredups". In some Linux systems, this is set by default to "ignoreboth" which covers both of the previous examples. This means that "ls" will not be saved, but "ls" would be saved by history. Adversaries can abuse this to operate without leaving traces by simply prepending a space to all of their terminal commands.

On Windows systems, the `PSReadLine` module tracks commands used in all PowerShell sessions and writes them to a file

(`$env:APPDATA\Microsoft\Windows\PowerShell\PSReadLine\ConsoleHost_history.txt` by default). Adversaries may change where these logs are saved using `Set-PSReadLineOption -HistorySavePath {File Path}`. This will cause `ConsoleHost_history.txt` to stop receiving logs. Additionally, it is possible to turn off logging to this file using the PowerShell command `Set-PSReadlineOption -HistorySaveStyle SaveNothing`.^{[2][3]}

Adversaries may also leverage a [Network Device CLI](#) on network devices to disable historical command logging (e.g. `no logging`).

ID: T1690

Sub-techniques: No sub-techniques

① **Tactic:** [Defense Impairment](#)

① **Platforms:** ESXi, Linux, Network Devices, Windows, macOS

Contributors: Austin Clark, @c2defense; Emile Kenning, Sophos; Vikas Singh, Sophos

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Procedure Examples

ID	Name	Description
G0082	APT38	APT38 has prepended a space to all of their terminal commands to operate without leaving traces in the HISTCONTROL environment. ^[4]
C0046	ArcaneDoor	ArcaneDoor included disabling logging on targeted Cisco ASA appliances. ^{[5][6]}
S1161	BPFDoor	BPFDoor sets the <code>MYSQL_HISTFILE</code> and <code>HISTFILE</code> to <code>/dev/null</code> preventing the shell and MySQL from logging history in <code>/proc/<PID>/environ</code> . ^[7]
S9015	BRICKSTORM	BRICKSTORM has impaired command logging through the use of <code>dev/null</code> which prevents generating output from the command and does not wait for input. ^[8]
S1186	Line Dancer	Line Dancer can disable syslog on compromised devices. ^[5]
G1051	Medusa Group	Medusa Group has removed PowerShell command history through the use of the PSReadLine module by running the PowerShell command <code>Remove-Item (Get-PSReadlineOption).HistorySavePath</code> . ^[9]
C0056	RedPenguin	During RedPenguin , UNC3886 used malware to clear the <code>HISTFILE</code> environmental variable and to inject into Junos OS processes to inhibit logging. ^{[10][11]}
G1041	Sea Turtle	Sea Turtle unset the Bash and MySQL history files on victim systems. ^[12]
S0692	SILENTTRINITY	SILENTTRINITY can bypass ScriptBlock logging to execute unmanaged PowerShell code from memory. ^[13]
S9024	SPAWNCHIMERA	SPAWNCHIMERA has disabled logging and log forwarding on Ivanti devices targeting the <code>dslogserver</code> process. ^{[14][15][16][17]}
G1048	UNC3886	UNC3886 has tampered with and disabled logging services on targeted systems. ^[18]
S1217	VIRTUALPITA	VIRTUALPITA can impair logging by setting the <code>HISTFILE</code> environmental variable to <code>0</code> and stopping the <code>vmsyslogd</code> service. ^[1]

Mitigations

ID	Mitigation	Description
M1039	Environment Variable Permissions	Prevent users from changing the <code>HISTCONTROL</code> , <code>HISTFILE</code> , and <code>HISTFILESIZE</code> environment variables. ^[19]
M1028	Operating System Configuration	Make sure that the <code>HISTCONTROL</code> environment variable is set to "ignoredups" instead of "ignoreboth" or "ignorespace".

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0563	Detection Strategy for Defense Impairment via Prevent Command History Logging across OS platforms.	AN1555	Detection of environment variable tampering (HISTFILE, HISTCONTROL, HISTFILESIZE) and absence of expected bash history writes. Correlation of unset or zeroed history variables with active shell sessions is indicative of adversarial evasion.
		AN1556	Detection of bash/zsh history suppression via HISTFILE/HISTCONTROL manipulation and absence of ~/.bash_history updates. Observing environment variable changes tied to terminal processes is a strong indicator.
		AN1557	Detection of PowerShell history suppression using Set-PSReadLineOption with SaveNothing or altered HistorySavePath. Correlating these options with PowerShell usage highlights adversarial evasion attempts.
		AN1558	Detection of unset HISTFILE or modified history variables in ESXi shell sessions. Correlation of suspicious shell sessions with no recorded commands despite active usage.
		AN1559	Detection of CLI commands that disable history logging such as 'no logging'. Anomalous lack of new commands in session logs while activity persists is a strong signal.

References

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